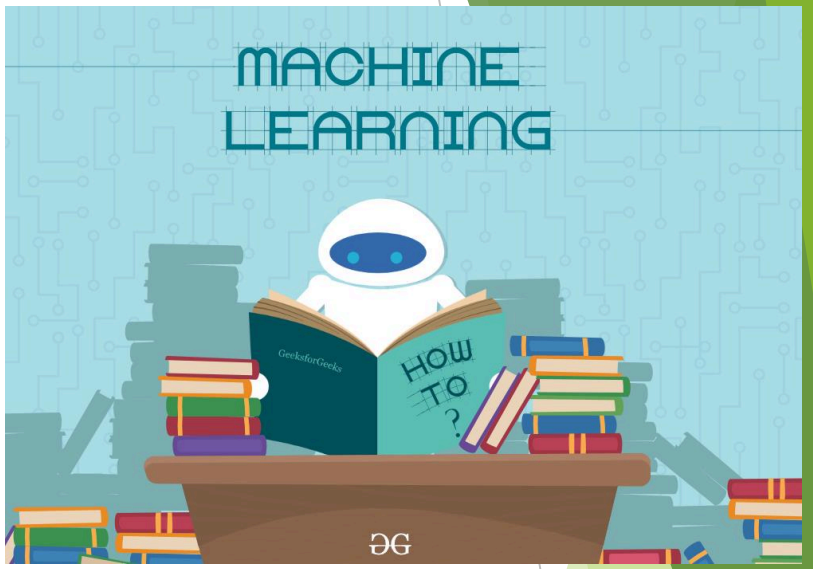
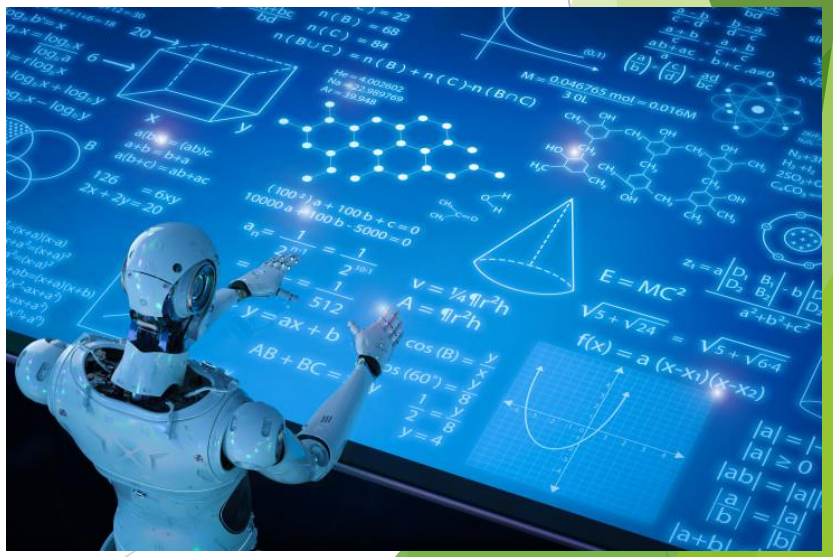
Machine Learning Basics: An Overview

Main Sources: “Machine Learning” by Tom Mitchell and “Artificial Intelligence – A Modern Approach” by Russell & Norvig and “Machine Learning: The Art & Science of Algorithms that make sense of Data” by Peter Flach

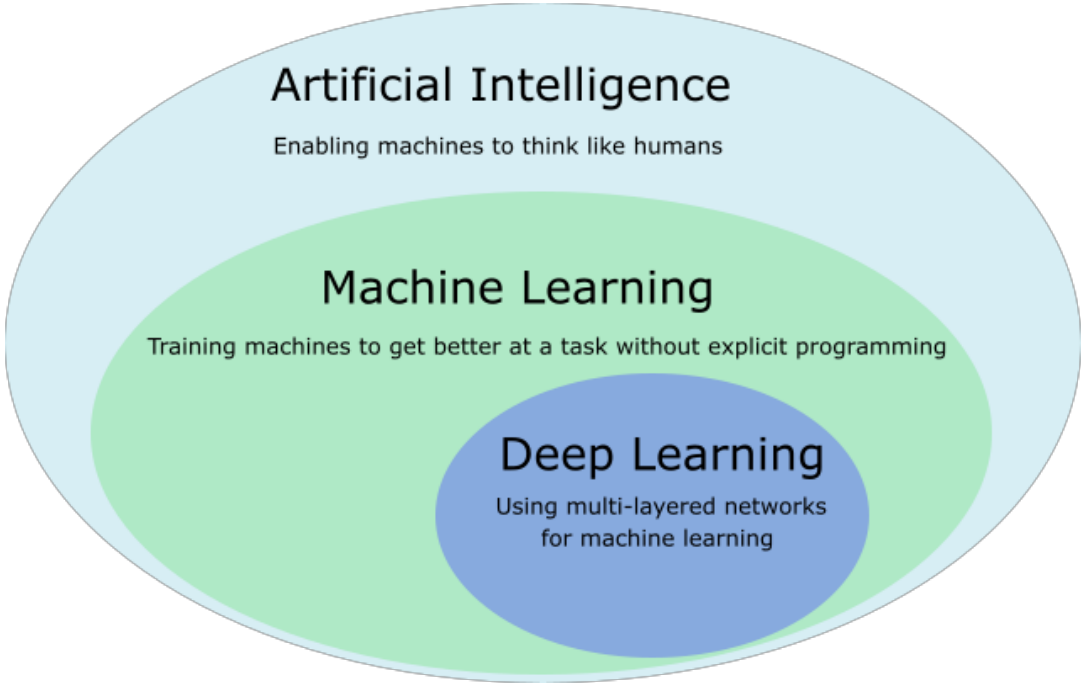
What is Machine Learning (ML)?

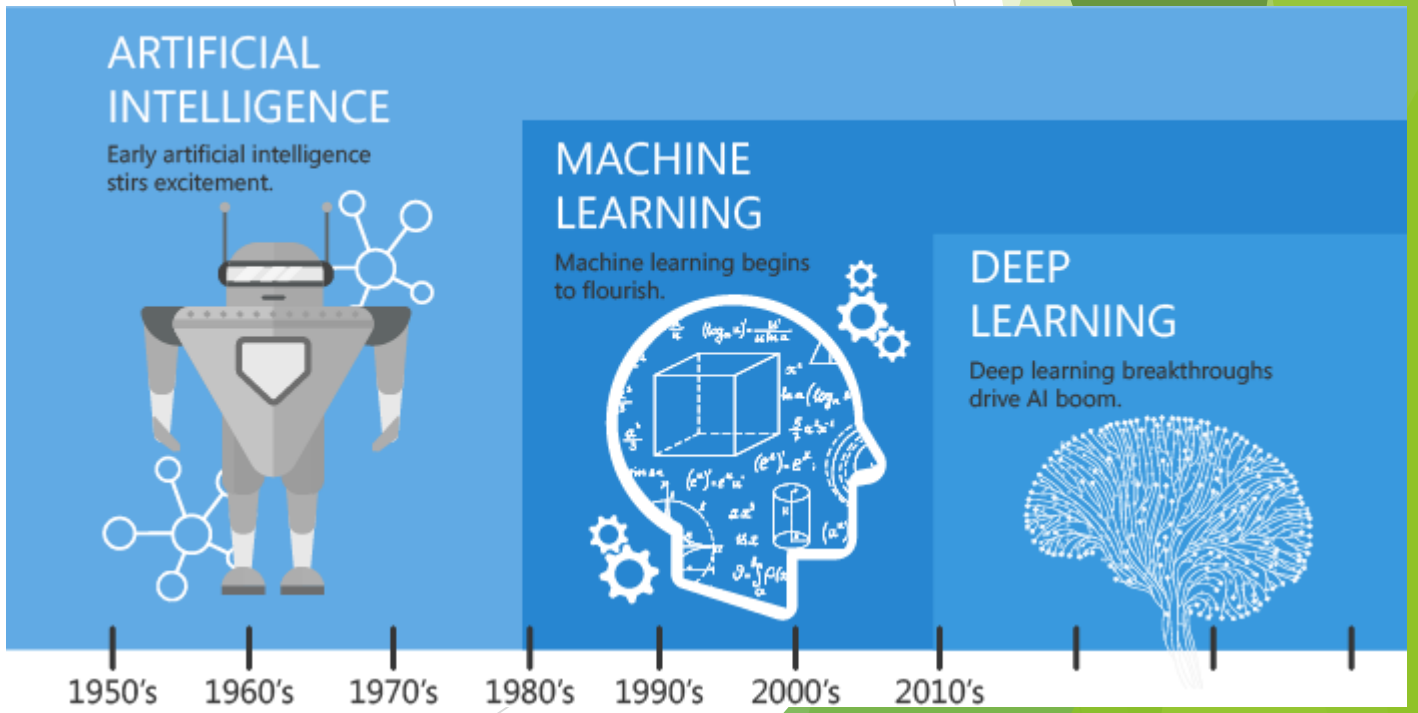
- As the name implies: ML is making machines learn analogous to humans
- Scientific study of algorithms & statistical models that computers use to perform a specific task without using explicit instructions, relying on patterns & inference (Source: Wiki)
- A field of artificial intelligence (AI) that provides systems the ability to learn and improve from experience automatically
- ML is data driven: leads to Data Science





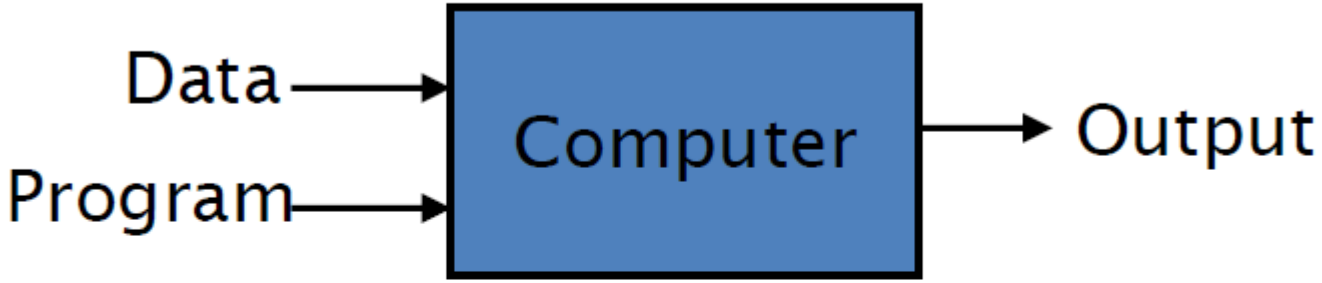
The Big Picture: AI & Machine Learning



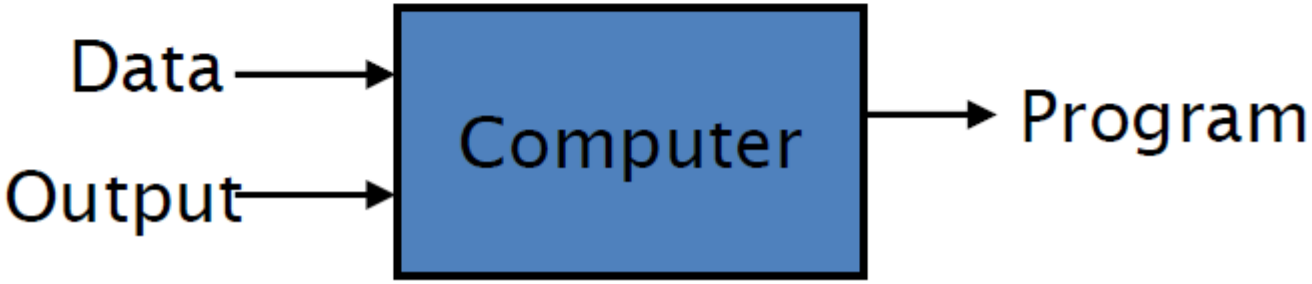


Traditional Programming vs Machine Learning

Traditional Programming

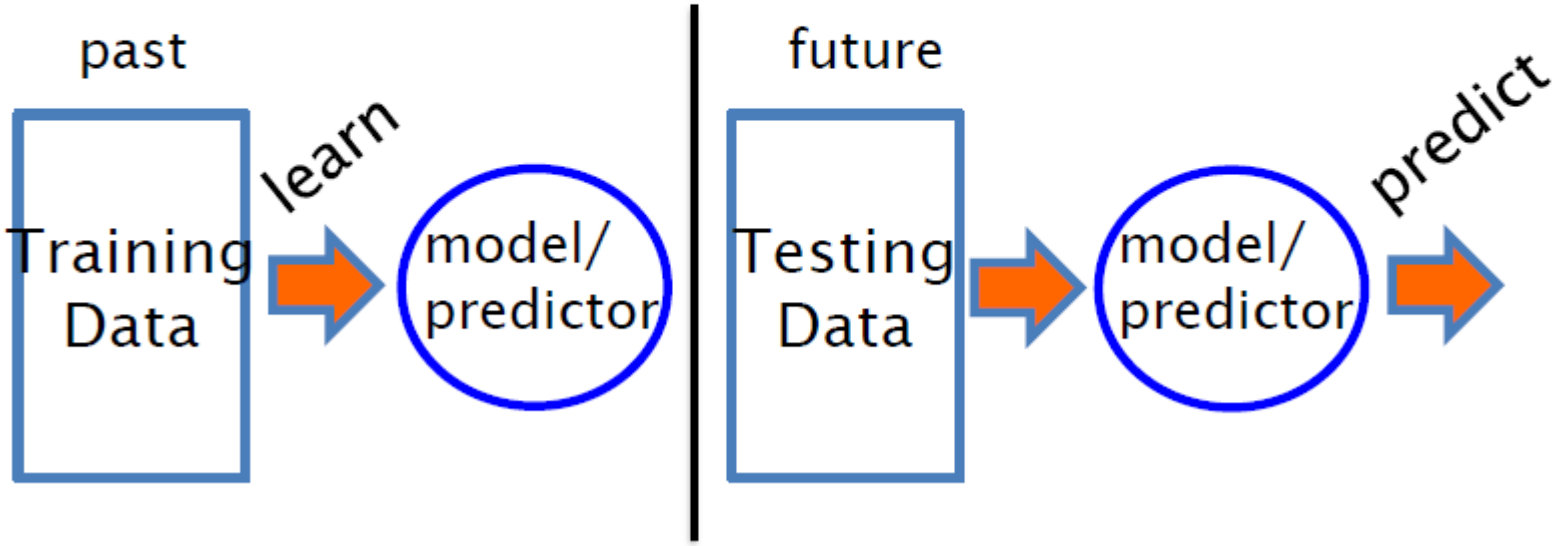


Machine Learning



Machine Learning & Predictive Analysis

Machine learning is about predicting the future based on the past.



Machine Learning, aka

data mining: machine learning applied to “databases”, i.e. collections of data

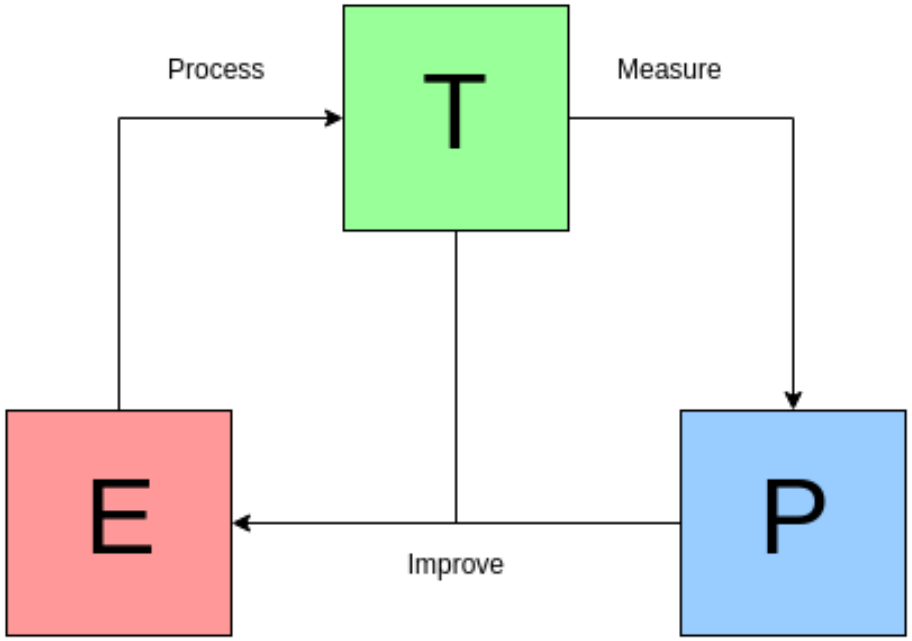
inference and/or *estimation* in statistics

pattern recognition in engineering

Formal Definition of Machine Learning

- As a broad subfield of artificial intelligence, machine learning is concerned with the design and development of algorithms and techniques that allow computers to "learn”

- A computer program is said to learn
  - from experience E
  - with respect to some class of tasks T
  - and performance measure P
  - if its performance at tasks in T
  - as measured by P
  - improves with experience E



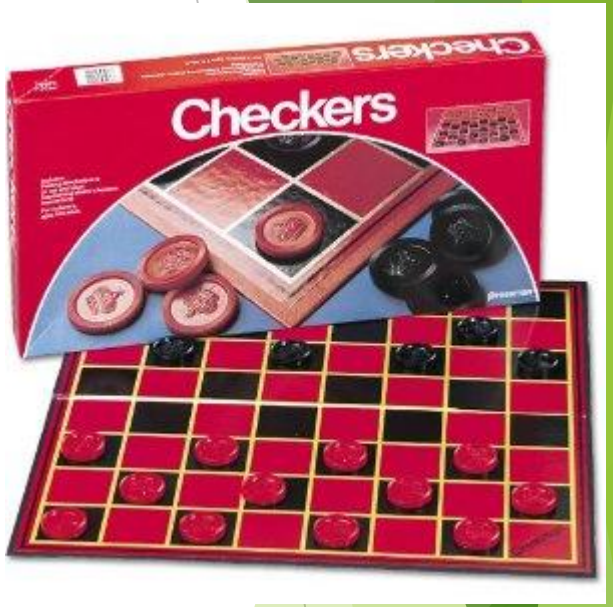
Source: Tom Mitchell, Machine Learning

Examples of ML Problems

- Checkers Playing
- Handwriting Recognition
- Robot Driving

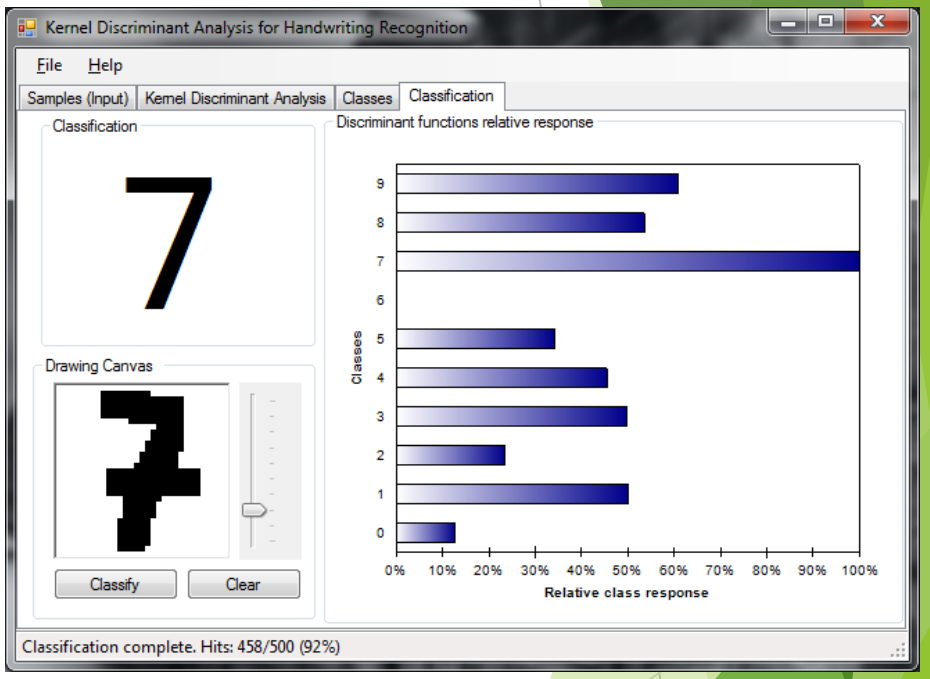
A Checkers Learning Problem

- Task T: Playing checkers
- Performance Measure P: Percentage of games won against opponents
- Training Experience E: Playing practice games against itself



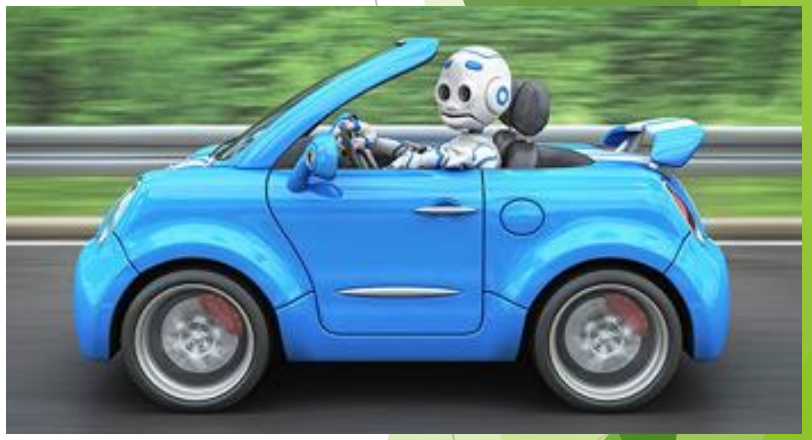
A Handwriting Recognition Problem

- Task T: Recognizing and classifying handwritten words within images
- Performance Measure P: Percentage of words correctly classified
- Training Experience E: A database of handwritten words with given classification



A Robot Driving Problem

- Task T: Driving on public four lane highways using vision sensors
- Performance Measure P: Average distance traveled before an error occurs (as judged by human overseer)
- Training Experience E: A sequence of images and steering commands recorded while observing a human driver



Data in Machine Learning

- Training Set
  - Data provided for learning the hypothesis
  - Used during the learning phase
- Test Set
  - Data on which the learned hypothesis is executed to evaluate its effectiveness
  - Used after the learning phase
- Validation Set
  - Additional data used for learning in order to avoid the possibility of the hypothesis overfitting the training data (In some situations, part of the training set can be kept aside for validation)
  - Used during the learning phase

Paradigms of Machine Learning

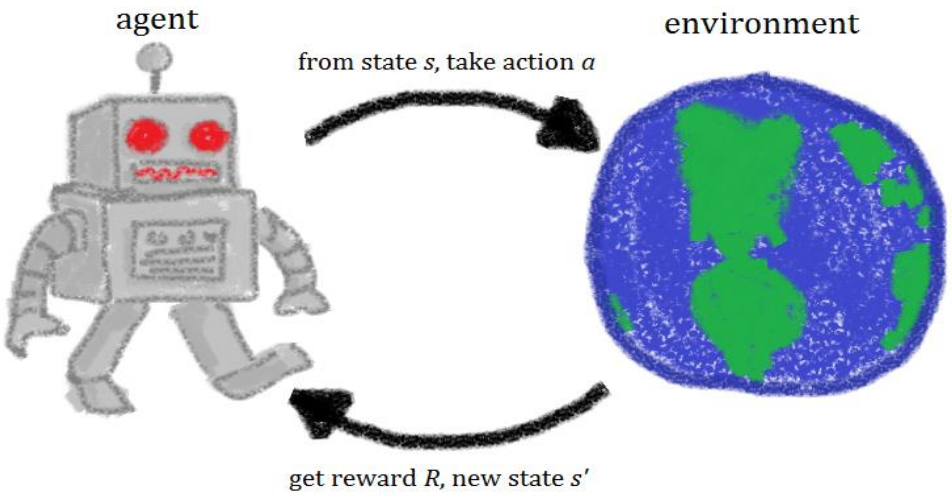
- Based on the data provided
  - Supervised Learning
  - Unsupervised Learning
  - Semi-supervised Learning
  - Reinforcement Learning

- Based on the process used in learning
  - Inductive Learning
  - Deductive Learning

- Based on when the learning occurs
  - Eager Learning
  - Lazy Learning

Supervised/Unsupervised/ Reinforcement Learning

- In **supervised learning** the class label of each training sample and the number of classes is known in advance, thus it is learning by examples
- In **unsupervised learning** there is no such prior knowledge, the learning is more intuitive, so it is learning by observation
- In **semi-supervised learning** there may be a training set with partial labels provided, this can be used to learn & add more labeled examples
- In **reinforcement learning**, actions need to be taken in an environment so as to maximize cumulative reward, thus there are no labeled samples but there are rewards to guide the learning

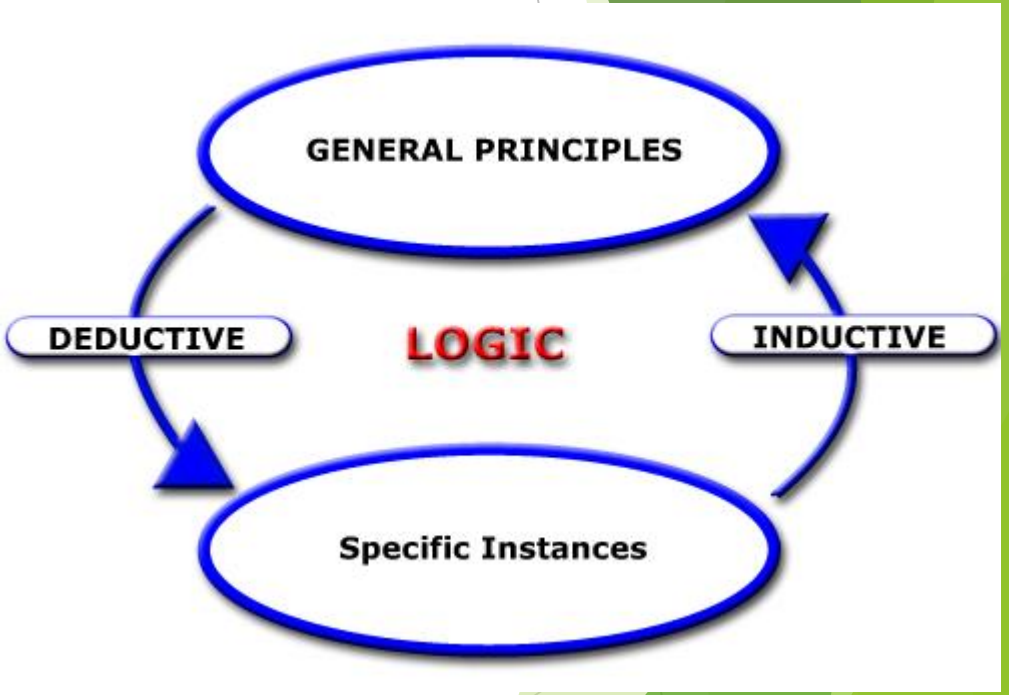


Reinforcement Learning

Inductive / Deductive Learning

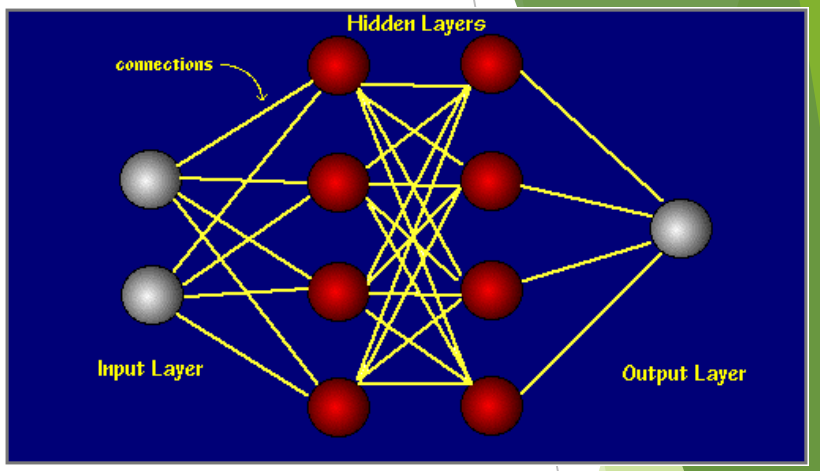
- **Inductive Learning** examines specific information, perhaps many pieces of specific information, to derive a general principle. Induction is employed
  - For example, in using specific propositions such as “This ice is cold”
  - And to infer general propositions such as “All ice is cold”

- **Deductive Learning** applies general principles to reach specific conclusions
  - For example, “All apples are fruit”
  - And “Some apples are red”
  - Therefore “Some fruit is red”



Eager / Lazy Learning

- **Eager Learning**
  - The technique learns a hypothesis over training data without waiting for test data or new unseen samples, e.g., Neural Networks

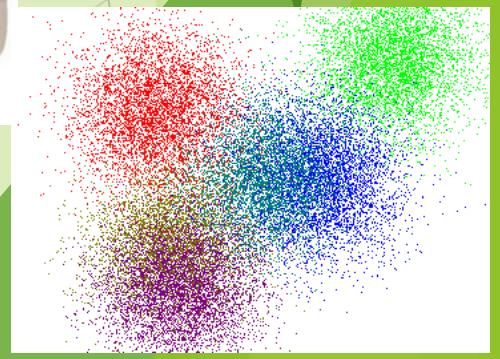


- **Lazy Learning**
  - The technique waits for a new sample to be provided in order to use existing data to learn the hypothesis, e.g., Case Based Reasoning (CBR)

Soft-CBR: Integration of Soft-Computing techniques to promote the CBR

Examples of ML Techniques

- Classification
  - Classify a target categorically: Decision Trees, Bayesian Classifiers, Neural Networks, Case Based Reasoning, k-Nearest Neighbors, Support Vector Machines, Genetic Algorithms
- Prediction
  - Predict the numeric value of a variable: Linear Regression, Multiple Linear Regression, Polynomial Regression
- Association Rule Mining
  - Learning Associations A => B (A implies B)
  - Apriori Algorithm is very popular here
- Cluster Analysis
  - Grouping objects based on similarity
  - k-means, COBWEB



Evaluating the Performance of ML Techniques

- Training Set Method
- Test Set Method
- k-holdout Strategy
- n-fold cross validation (cv)

Using the Training Set for Evaluation

- After learning the hypothesis, execute it over the training set itself to check for performance in terms of parameters such as accuracy
- Advantage: More availability of data for evaluation
- Disadvantage: Overfitting, i.e., learned hypothesis may be too specific to the training data provided, not generic enough

Using a Distinct Test Set

- Keep a separate test set that is completely distinct from the training set and evaluate accuracy over the test set to ensure generality of hypothesis
- Advantage: Avoids overfitting and makes learning more generic
- Disadvantage: Lack of availability, as distinct test sets may be harder to find

Using the k-holdout strategy

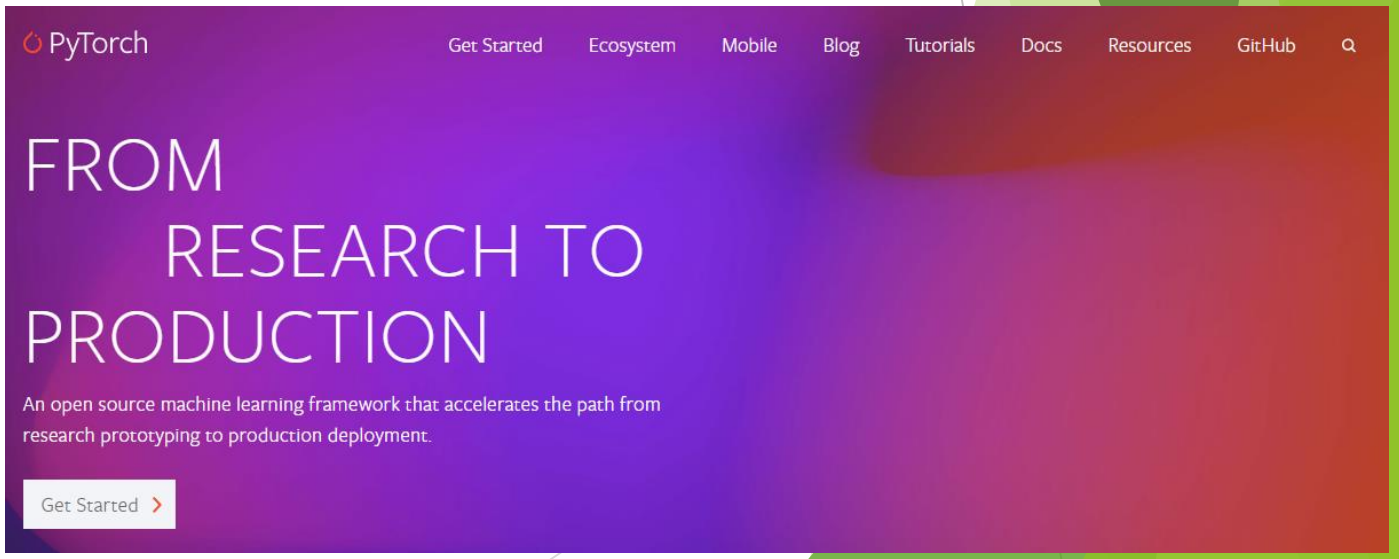
- Set aside k samples from the training set for testing
- Advantage: overcomes the drawbacks of the previous two methods
- Disadvantage: Since some of the training set is used for testing, it indirectly reduces the size of the training set

Using n-fold cross validation (cv)

- Consider the given data, make n folds or parts, used (n-1)/n folds for training and the remaining for testing, repeat this n times for different ranges
- E.g., if the data has 100 samples, and 4-fold cv is used, then use 75 samples (3/4) for training and the remaining 25 samples (1/4) for testing and repeat this 4 times with different ranges each time
- Advantage: provides a distinct training & test set to avoid overfitting and also makes the learning even more generic by repeating multiple times
- Disadvantage: Consumes more time and increases the complexity of evaluation
- Note: This is usually the most preferred method as it is found to give the best results in learning a generic hypothesis; most common values of n are 4 and 10

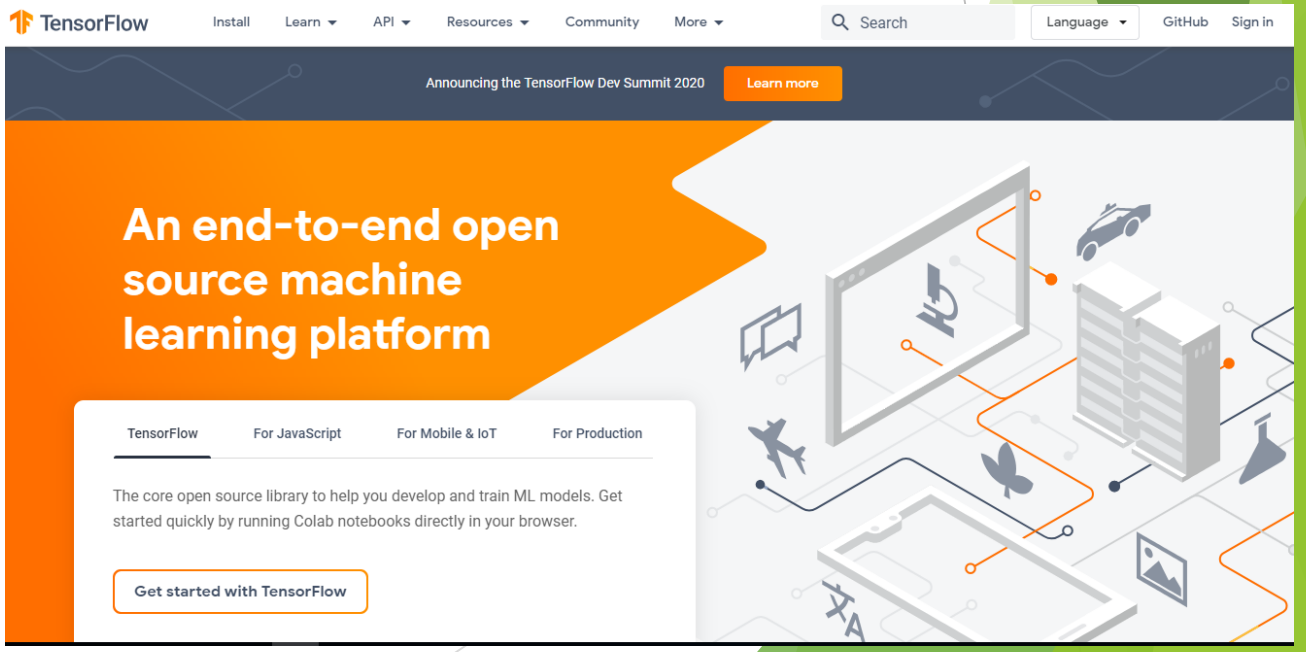
ML Software Tool: PyTorch

- PyTorch --- https://pytorch.org/
  - Free Open Source ML Library
  - Used a lot in Computer Vision and NLP (Natural Language Processing)
  - Developed at Facebook's AI Research Lab (FAIR)
  - Written in Python, C++, CUDA



ML Software Tool: TensorFlow

- TensorFlow --- https://www.tensorflow.org/install
  - Free Open Source Software Library & Symbolic Math Library
  - Used much for ML applications such as Neural Networks
  - Developed by Google Brain team at Google
  - Released under Apache
  - Written in Python, C++, CUDA



ML Software Tool: WEKA

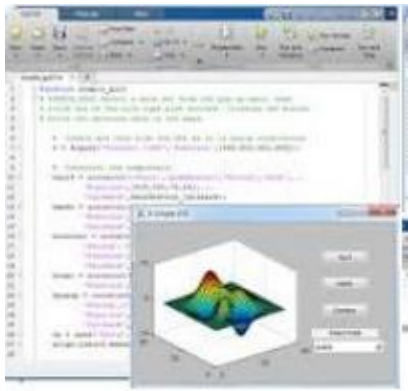
- WEKA --- https://sourceforge.net/projects/weka/
  - Waikato Environment for Knowledge Analysis
  - Free Software with implementations of ML algorithms
  - Used as a good source of Data Mining techniques
  - Developed at University of Waikato, New Zealand
  - Released under GNU General Public License
  - Written in Java

Other Relevant Software useful for ML

- R: Programming Language & Free Software Environment used in Statistical Computing
  - (https://www.r-project.org/)
- MATLAB (Matrix Laboratory): Proprietary Software for Numerical Computing
  - (https://www.mathworks.com/products/matlab-online.html)
- Apache Mahout: Distributed Scalable ML Algorithms, focusing on Linear Algebra, well-suited for Big Data
  - (https://mahout.apache.org/)

R

Programming language



MATLAB

Apache Mahout

www.educba.com

Resources: Selected Websites with ML Datasets

- **UCI Machine Learning Datasets: https://archive.ics.uci.edu/ml/datasets.html**
- Deep Learning Datasets: http://deeplearning.net/datasets/
- PASCAL Visual Object Classes: http://host.robots.ox.ac.uk/pascal/VOC/
- UCI KDD Archive: http://kdd.ics.uci.edu/summary.data.application.html
- Statlib: http://lib.stat.cmu.edu/
- Delve: http://www.cs.utoronto.ca/ delve/
- Weka: http://www.cs.waikato.ac.nz/ml/weka/

Resources: Selected ML, AI & related Conferences

- ICML: International Conference on Machine Learning
- AAAI: Conference on Artificial Intelligence (by AAAI: Association for Advancement of Artificial Intelligence)
- NIPS: Neural Information Processing Systems
- SIGKDD (or KDD): Knowledge Discovery and Data Mining (by ACM: Association for Computing Machinery)
- ICDM: International Conference on Data Mining (by IEEE: Institute of Electrical & Electronics Engineers)
- ECML/PKDD: European Conference on Machine Learning / Principles of Knowledge Discovery & Data Mining
- CVPR: Computer Vision & Pattern Recognition (by IEEE)
- CIKM: Conference on Information & Knowledge Management (by ACM)
- ICTAI: International Conference on Tools with Artificial Intelligence (by IEEE)
- IEEE Big Data: IEEE's International Conference on Big Data
- SIGMOD: Intl. Conf. on Management of Data (by ACM's Special Interest Group on Mgmt. of Data)
- ACL: Annual Conf. of the Association for Computational Linguistics
- IJCAI: International Joint Conference on Artificial Intelligence

Resources: Selected ML, AI & Related Journals

- Journal of Machine Learning Research (JMLR)
- Springer's Machine Learning
- Elsevier's Neural Networks
- Elsevier's Artificial Intelligence
- ACM Transactions on Knowledge Discovery and Data Mining (TKDD)
- IEEE Transactions on Neural Networks and Learning Systems
- Journal of Artificial Intelligence Research (JAIR)
- ACM SIGKDD Explorations (by Special Interest Group on Knowledge Discovery & Data Mining
- ACM SIGMOD Record (by Special Interest Group on Management of Data)
- Data Mining and Knowledge Discovery (DMKD)
- IEEE Intelligent Systems

Summary

- Definition of Machine Learning (ML)
- Examples of ML Problems
- Data in Machine Learning
- Types of Learning
- Examples of ML Techniques
- Evaluation Methods
- ML Resources: Tools, Websites, Conferences, Journals